CS4391 Introduction to Computer Vision
Project Presentation and Final Report

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April 23, 2026

1 Presentation

The project presentations will be held **on 5/5 and 5/7 in GR 2.302 between 11:30 AM and 12:45 PM.**

The assignments of the groups are

- 5/5, Tuesday, Groups: 1, 2, 3, 4, 5, 6, 7, 8, 10, 11
- 5/7, Thursday, Groups: 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23

Schedule:

- **11:30am - 11:40am:** teams set up robots and test demos (please test it beforehand, there is only 10 minutes to set up the demo)
- **11:40pm - 12:45pm:** presentations, each team has 5 minutes (which includes the demo time)

The list of projects:

- Group 1: ARMoR: Real-Time Human Pose Estimation on NVIDIA Jetson Orin Nano
- Group 2: CubeVision: Rubik's Cube State Reconstruction and Solving Using Computer Vision
- Group 3: F.O.R.G.E.: Fine-tuned Optical Reconstruction for Gesture Extraction
- Group 4: Autonomous Drone with Vision-Based Object Tracking
- Group 5: Real-time Visual Assistance Through Depth Estimation
- Group 6: Human-Object Interaction Detection: Recognizing Instantaneous Interaction Frames in Ball-Throwing Footage
- Group 7: Human Hand Pose Estimation
- Group 8: Real-Time Custom Object Detection and Tracking on NVIDIA Jetson Orin Nano

- Group 10: Real-Time American Sign Language Translation Via YOLO26 and MediaPipe-Assisted Sequence
- Group 11: StockSync: Automated Retail Inventory Tracking
- Group 12: Hasta Detection Program: Real-Time Recognition of Bharatanatyam Mudras
- Group 13: Monocular Depth Estimation via Fine-Tuning Depth Anything V2 on Indoor and Outdoor Scenes
- Group 14: Self-Driving RC Car Built on Camera-Based Object Detection
- Group 15: SpineSense: A Real-Time Multi-Label Posture Detection and Scoring System
- Group 16: Real-Time Semantic Segmentation of Indoor Scenes on Jetson Orin Nano
- Group 17: Real-Time Plate Food Detection on Jetson Orin Nano
- Group 19: I Can't See: Custom Object Detection
- Group 20: Semantic Segmentation of Indoor Scenes
- Group 21: Real-Time Package Theft Alert System
- Group 22: Increasing the Accuracy of Visual Grounding
- Group 23: Pose-Based Red Light, Green Light Using Embedded Computer Vision

Each group has 5 minutes for the presentation, demo and questions. Please use slides to describe your project, and show a demo of the project.

Upload your slides to the following box link before your presentation. We will use one laptop to present: `https://utdallas.app.box.com/f/137762ec49274e48bc619991c9c8374b`

Make sure you practice your presentation beforehand. A timer will be used. You will have to stop the presentation if you run over 5 minutes.

Evaluation criteria: The grading will be based on the overall quality of the presentation in terms of content, clarity, and question answering.

2 Final Report

The project final report should be prepared using the CVPR latex template. A useful online LaTex tool is Overleaf `https://www.overleaf.com/`. We have the CVPR latex template accessible here via overleaf: `https://www.overleaf.com/read/gpjssbtrrpqm`. You can download a copy of the template or make a copy in overleaf for your own project, and then edit it.

In this project final report, please describe the following items according to your project:

- **Title**. The title of your project.
- **Team Members**. List the names of the team members.
- **Abstract**. Give an overview of the project.

- **Introduction**. Describe the motivation of the project, i.e., why do you want to work on this problem. Then describe an overview of the framework/method/system.
- **Related Work**. Discuss the related work of your project.
- **Method**. Describe your solution for the project. For example, describe each component of the framework in details. Try to use figures to illustrate the method instead of only using text. "A picture is worth a thousand words".
- **Experiments**. In this section, you can first describe the task and evaluation metrics. Then describe what experiments you have done for the project by adding experimental results to the report. Use figures and plots to show these results.
- **Conclusion**. Describe the take-home messages of the project and conclude the report.
- **References**. Cite related works in the report.

Evaluation criteria: The grading will be based on the overall quality of the report in terms of writing, content and clarity. Please make sure that you have the following components in your project:

- Data collection using the real camera
- Data labeling if supervision is needed
- Model training with the labeled data
- Model optimization using tensorRT
- Model deployment to Jetson Nano Orin
- Model evaluation on datasets

Minimum page requirement: 4 pages. The report should be at least 4 pages with the CVPR format (excluding references, i.e., without references, the content should be at least 4 pages). You can go beyond 4 pages, but make sure it is less than 6 pages (excluding references).

An example CVPR paper: you can check the structure of the following paper for reference `https://yuxng.github.io/Papers/2021/chao_cvpr21.pdf`.

3 Project Submission

Please submit the following items to eLearning. You can zip all the files.

- (Required) Final report in pdf format
- (Required) Presentation slides in pdf format (Do not have UNet IDs in it for releasing online)
- (Required) Source code of your project
- (Required) A demo video in mp4 format